



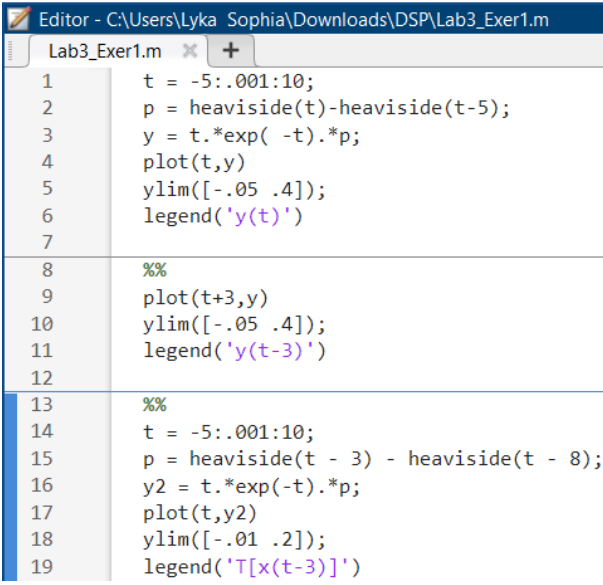
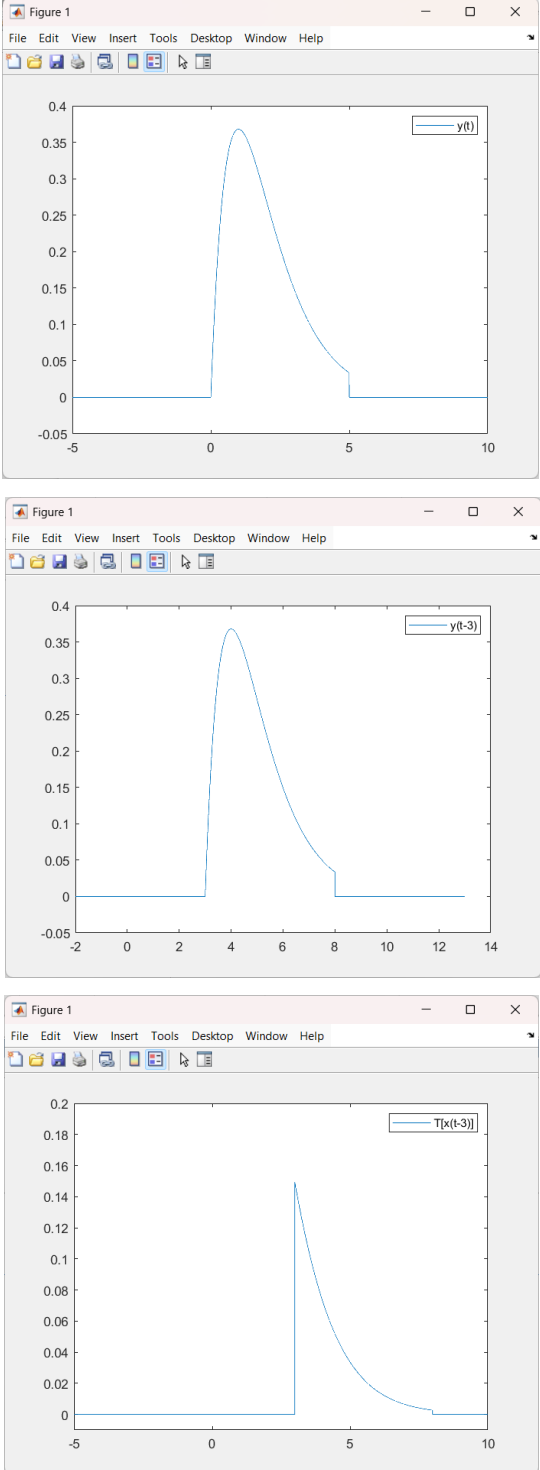
LINEARITY AND TIME-INVARIANCE

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Program:	BSCpE	Section:	A461

A. Guided Exercises:

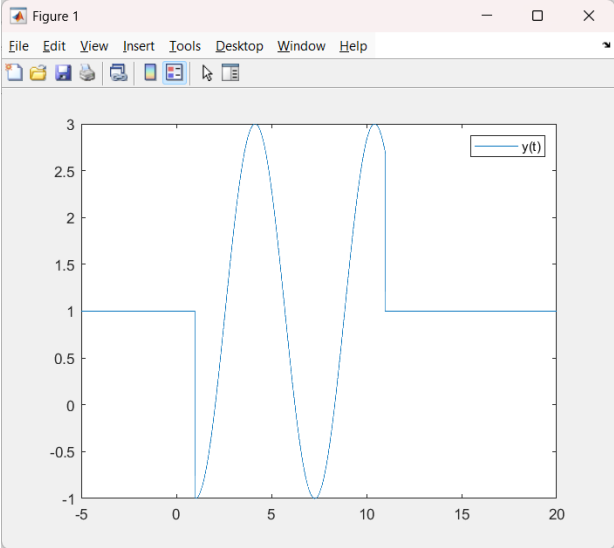
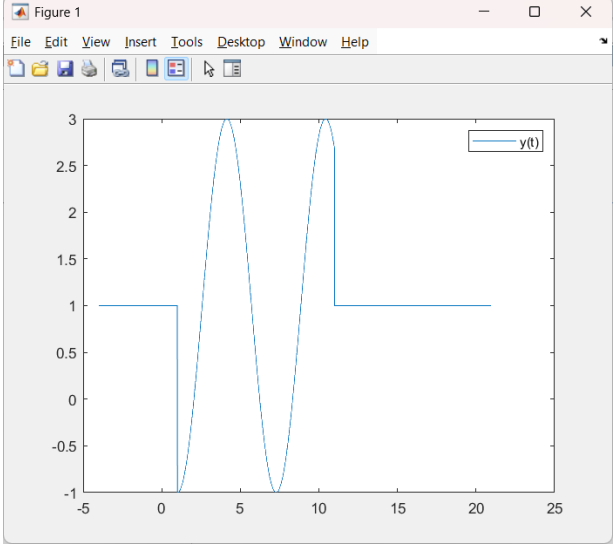
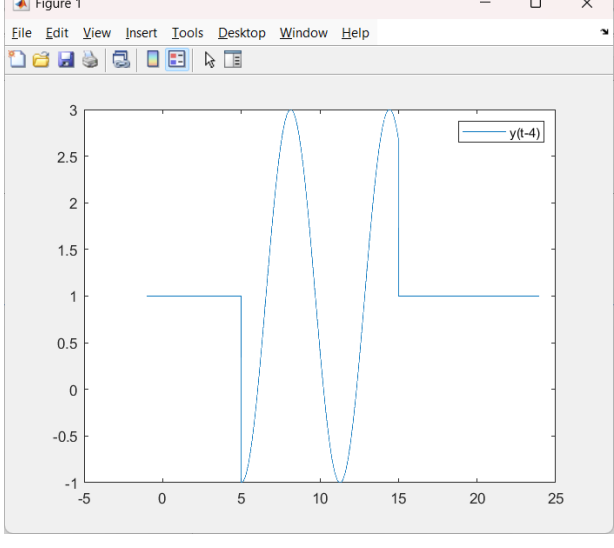
(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

Code and Execute the Solution in Example 1

Photo/s of MATLAB Codes	Photo/s of MATLAB Output
 <pre>1 t = -5:.001:10; 2 p = heaviside(t)-heaviside(t-5); 3 y = t.*exp(-t).*p; 4 plot(t,y) 5 ylim([-0.05 .4]); 6 legend('y(t)') 7 8 %% 9 plot(t+3,y) 10 ylim([-0.05 .4]); 11 legend('y(t-3)') 12 13 %% 14 t = -5:.001:10; 15 p = heaviside(t - 3) - heaviside(t - 8); 16 y2 = t.*exp(-t).*p; 17 plot(t,y2) 18 ylim([-0.01 .2]); 19 legend('T[x(t-3)]')</pre>	

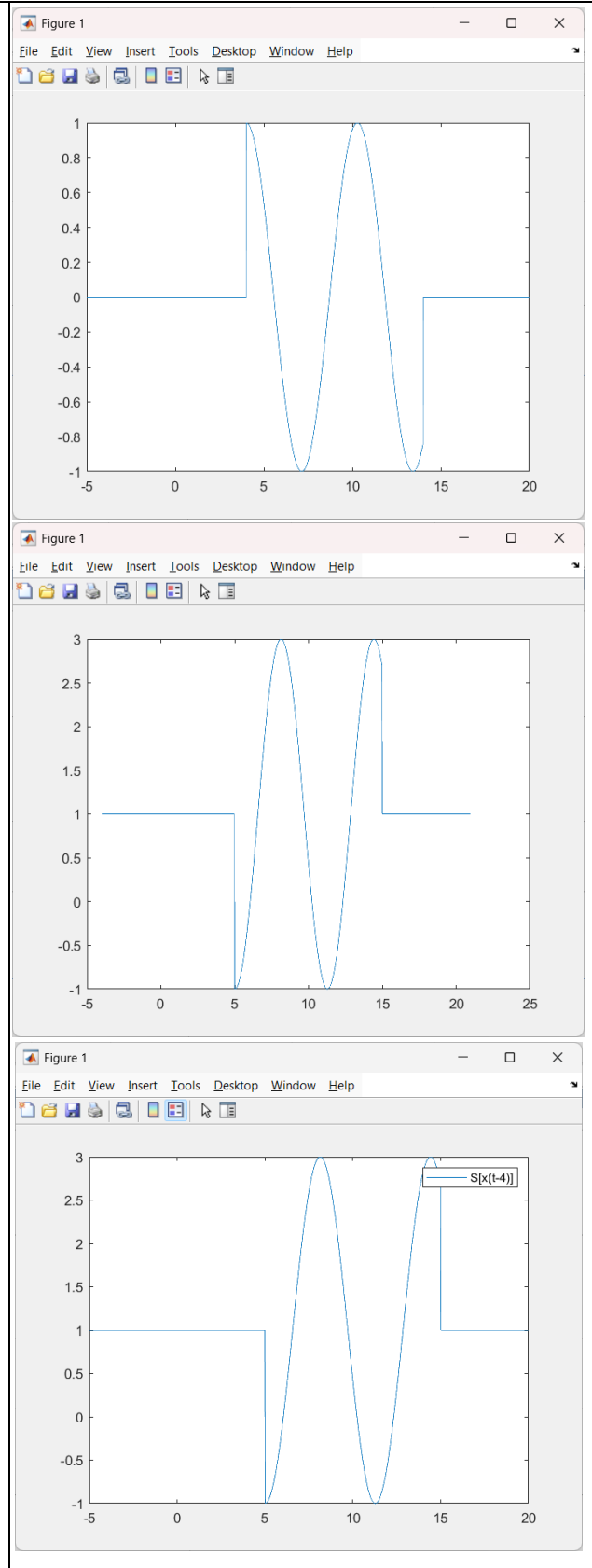


Code and Execute the Solution in Example 2

Photo/s of MATLAB Codes	Photo/s of MATLAB Output
<pre>Editor - C:\Users\Lyka Sophia\Downloads\DSP\Lab3_Exer2.m Lab3_Exer1.m Lab3_Exer2.m 1 t = -5:.001:20; 2 p = heaviside(t-1)-heaviside(t-11); 3 y = 1- 2*cos(t-1).*p; 4 plot(t,y) 5 legend('y(t)') 6 7 %% 8 p = heaviside(t)- heaviside(t-10); 9 x = cos(t).*p; 10 plot(t+1,1-2*x) 11 legend('y(t)') 12 13 %% 14 plot(t + 4,y) 15 legend('y(t-4)') 16 17 %% 18 p = heaviside(t - 4) - heaviside(t - 14); 19 x = cos(t - 4).*p; 20 plot(t,x) 21 22 %% 23 p = heaviside(t-4) - heaviside(t-14); 24 x = cos(t-4).*p; 25 plot(t+1,1-2*x) 26 27 %% 28 t = -5:.01:20; 29 p = heaviside(t-5)- heaviside(t-15); 30 y2 = 1-2*cos(t-5).*p; 31 plot(t,y2); 32 legend('S[x(t-4)]')</pre>	  



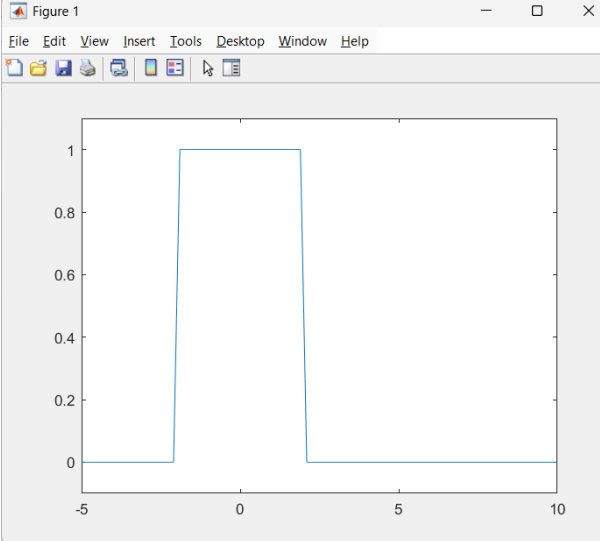
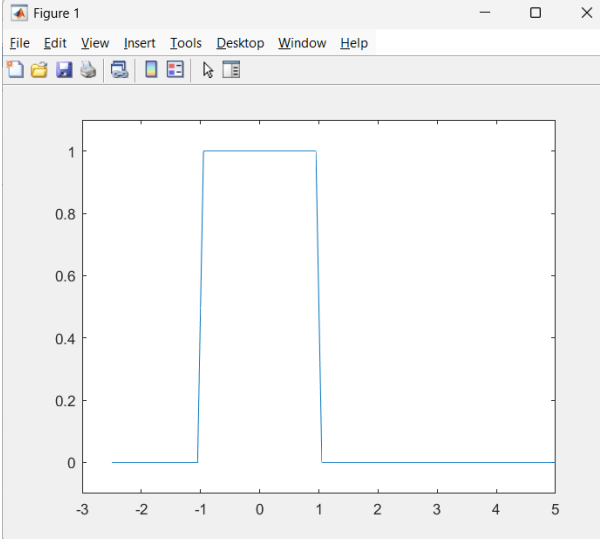
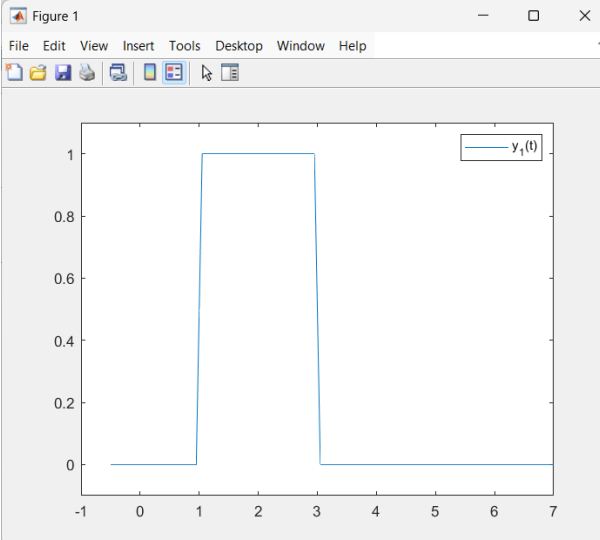
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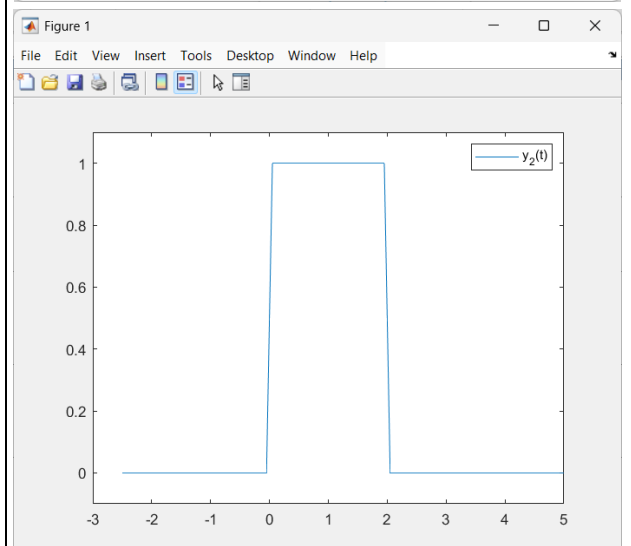
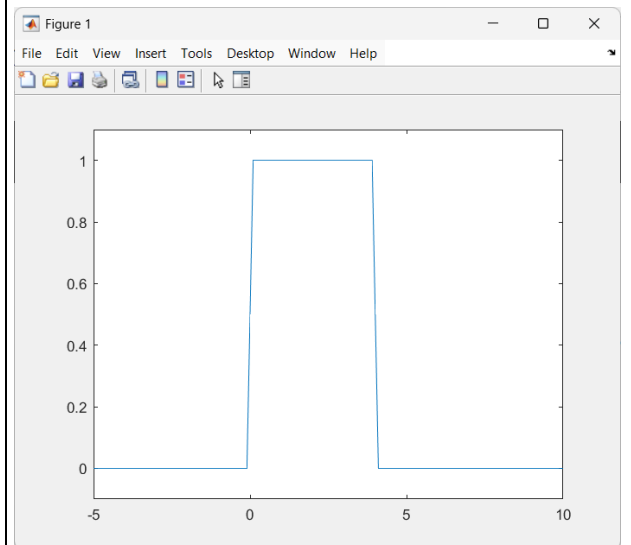
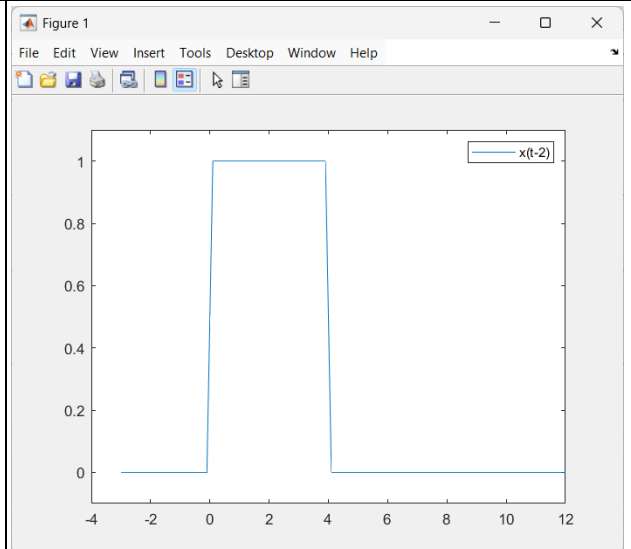
LINEARITY AND TIME-INVARIANCE

Code and Execute the Solution in Example 3

Photo/s of MATLAB Codes	Photo/s of MATLAB Output
<pre> 1 t = -5:1:10; 2 x = heaviside(t + 2) - heaviside(t - 2); 3 plot(t,x); 4 ylim([-1 1.1]) 5 6 %% 7 plot((1/2)*t,x); 8 ylim([-1 1.1]); 9 10 %% 11 plot((1/2) * t + 2,x); 12 ylim([-1 1.1]); 13 legend('y_1(t)') 14 15 %% 16 plot(t+2,x); 17 ylim([-1 1.1]); 18 legend('x(t-2)') 19 20 %% 21 t = -5:1:10; 22 x2=heaviside(t)- heaviside(t-4); 23 plot(t,x2); 24 ylim([-1 1.1]); 25 26 %% 27 plot((1/2) * t,x2); 28 ylim([-1 1.1]); 29 legend('y_2(t)') </pre>	  

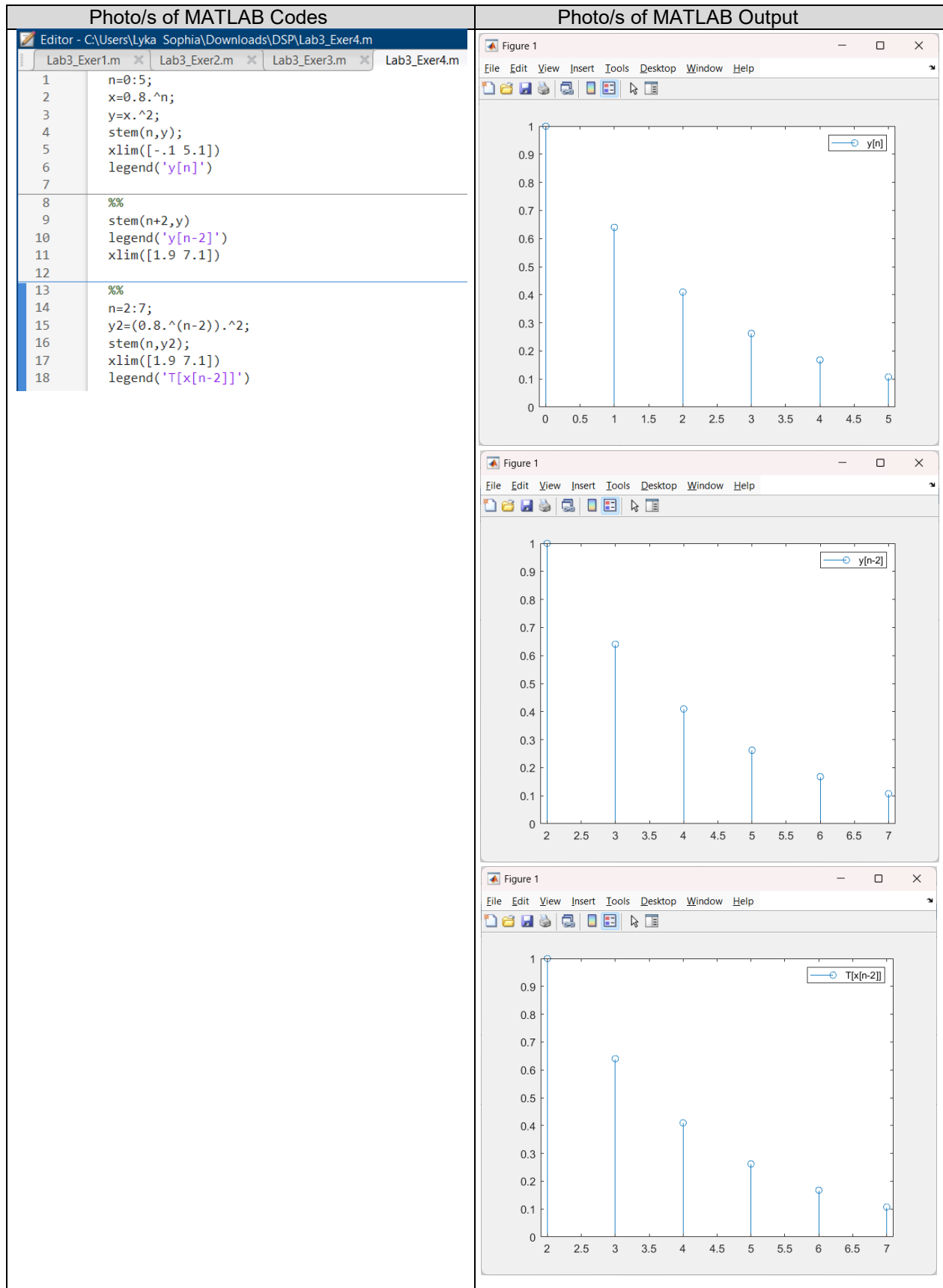


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Code and Execute the Solution in Example 4





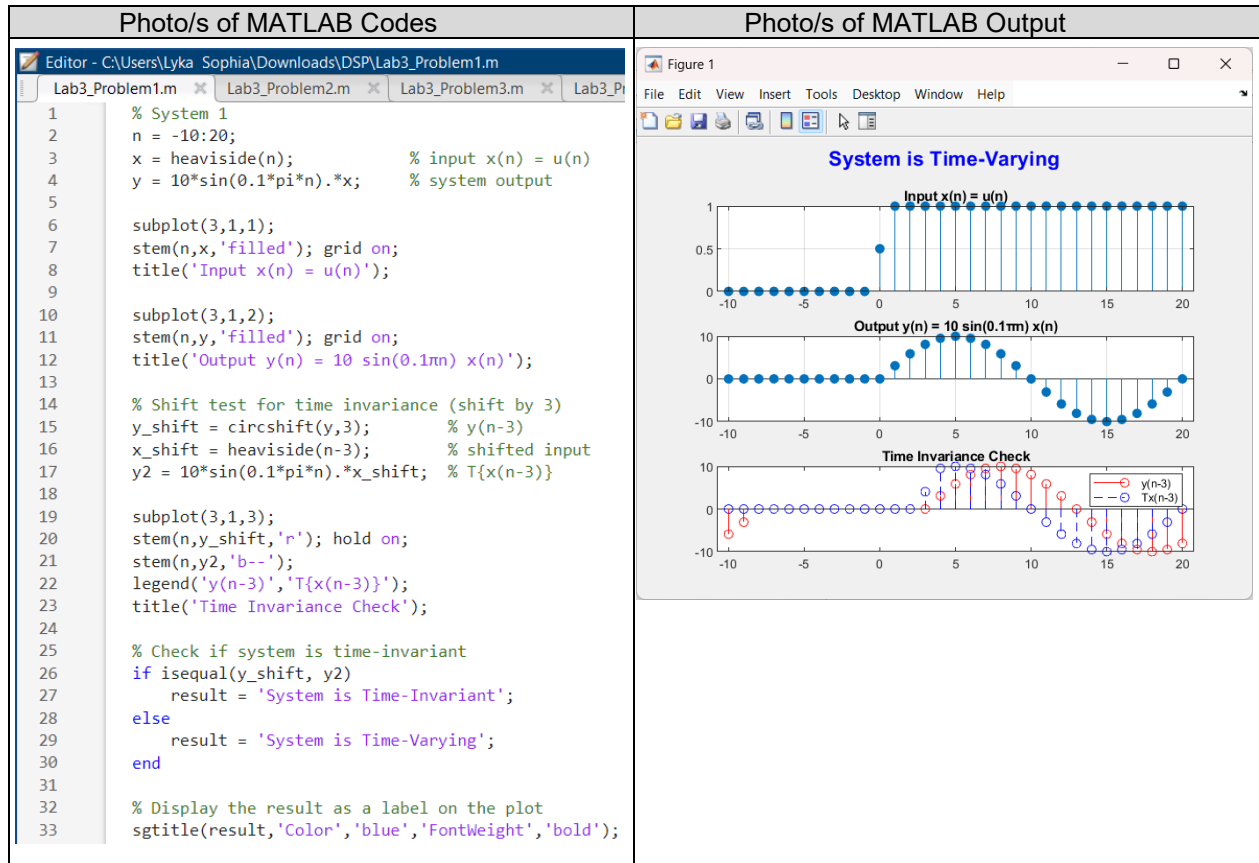
LINEARITY AND TIME-INVARIANCE

B. Problem Exercises:

(Please include the following: 1. Photos of the codes and 2. Outputs after executing the MATLAB code.)

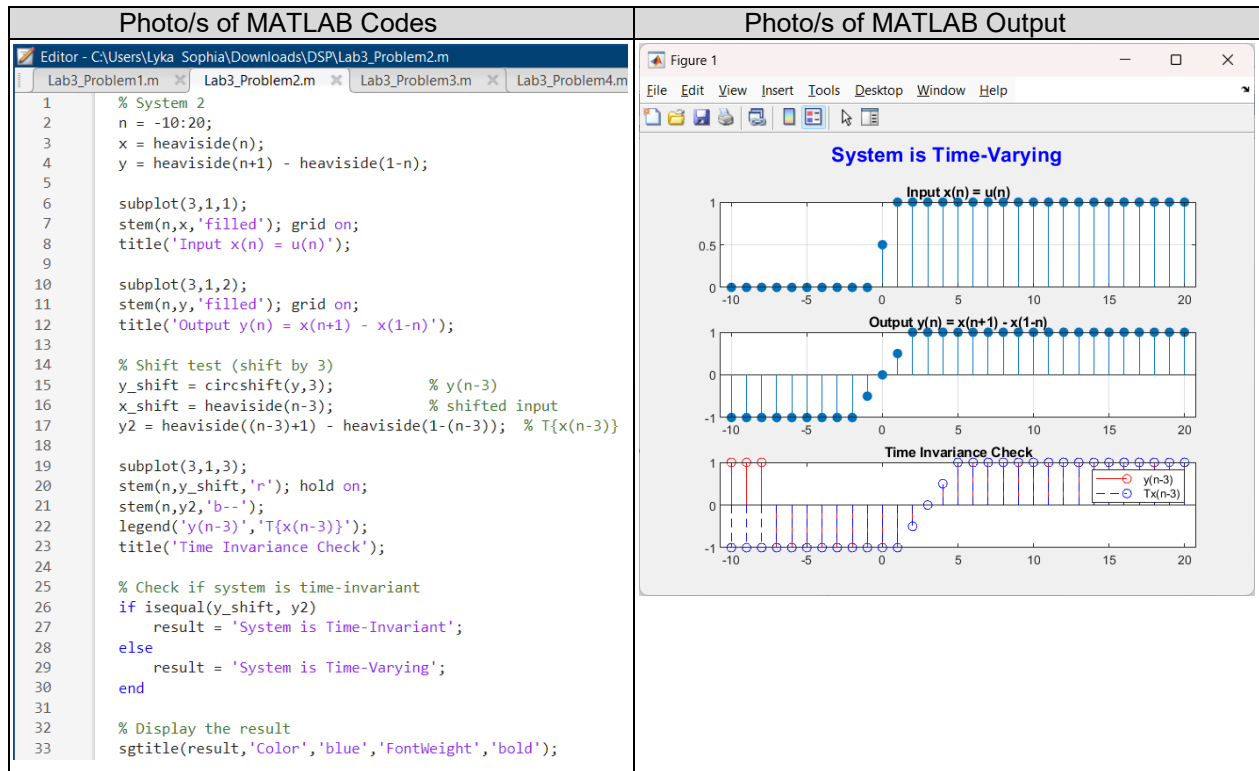
Determine the relationship (*time varying or time-invariant relationship*) of the given systems in the Lab Guide.

Problem 1:

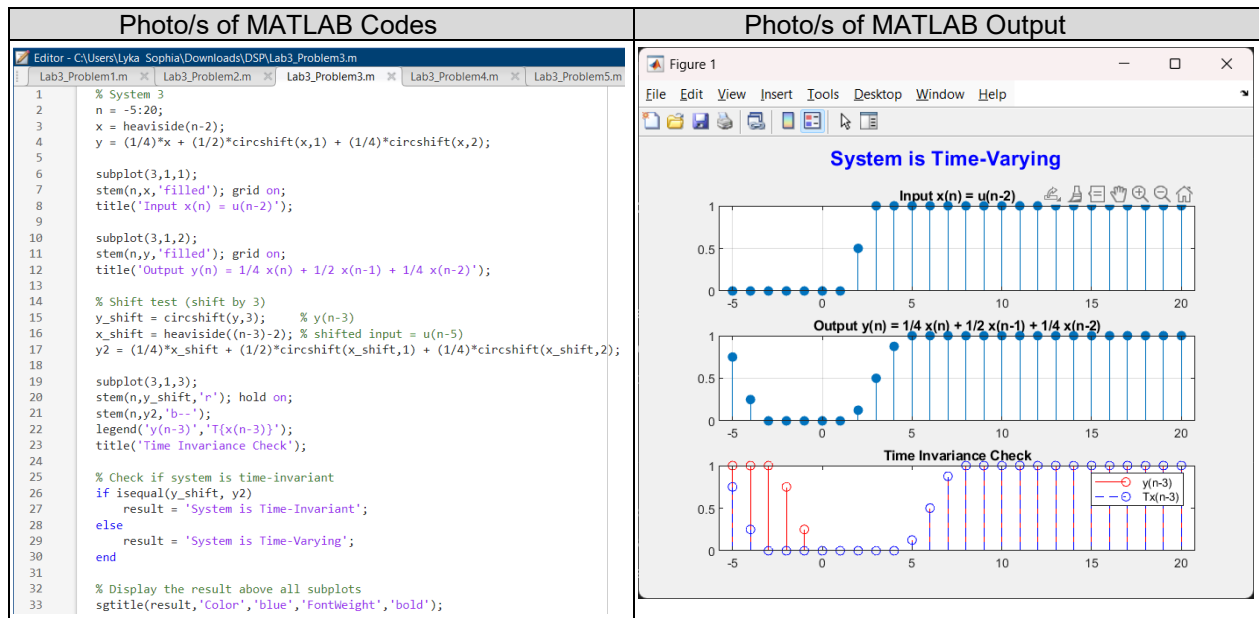




Problem 2:

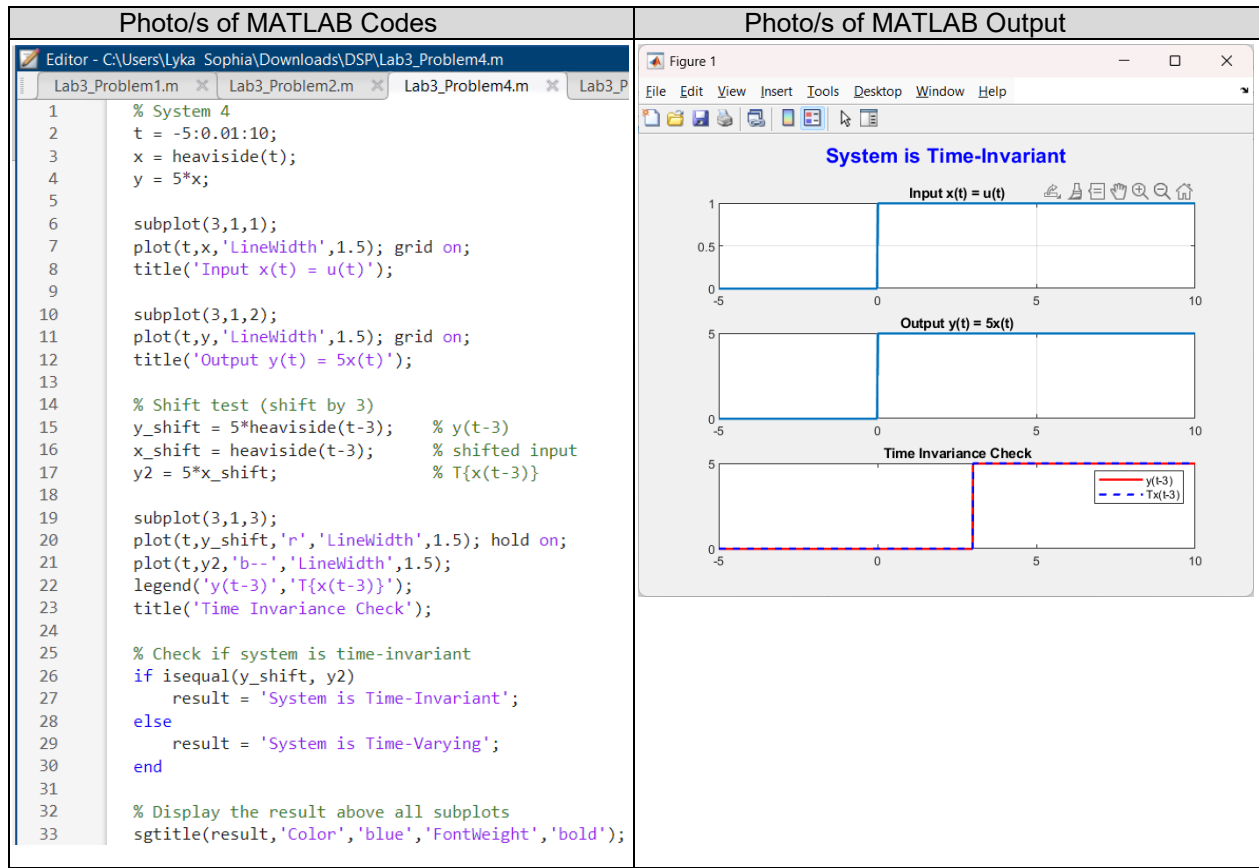


Problem 3:

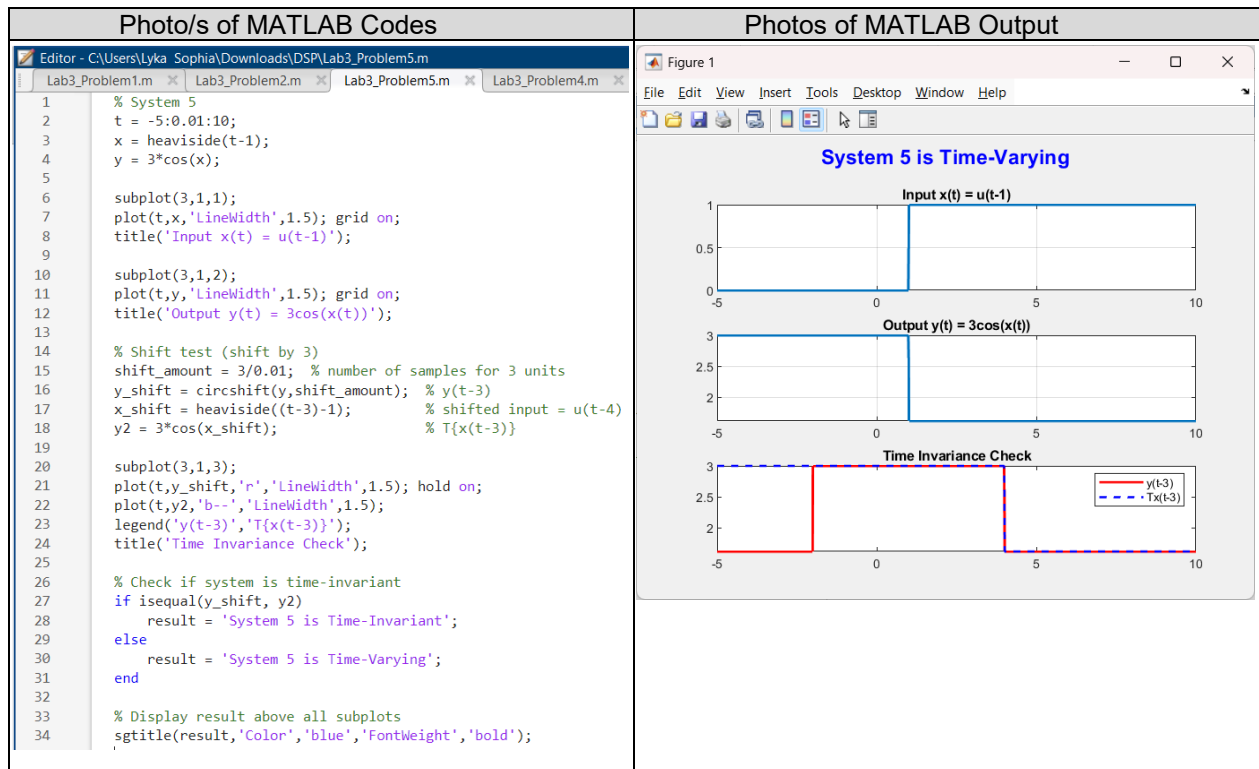




Problem 4:

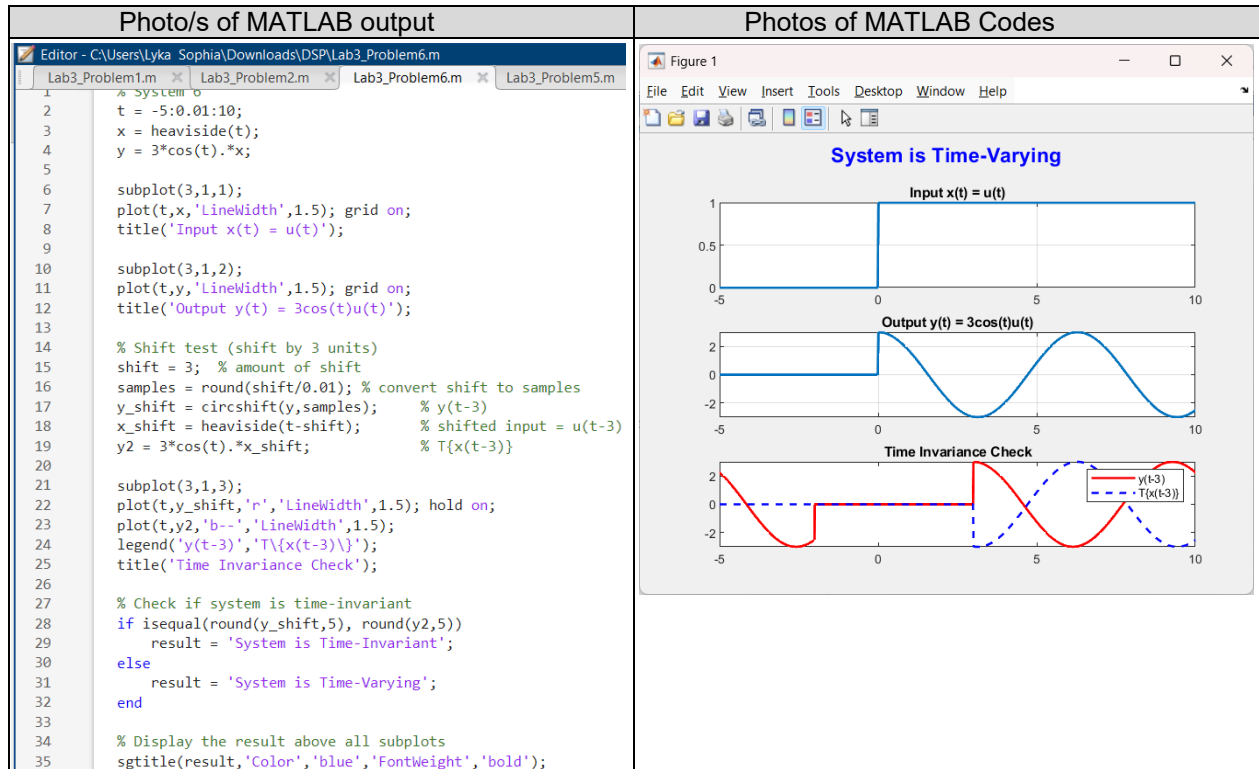


Problem 5:





Problem 6:



C. Findings ,Observations, and Comments:

Guiding Questions :

1. What did you learn in the guided exercises? Did you learn something new?
2. What are your findings, observations, and comments in the guided exercises?
3. What are your findings, observations, and comments in the problem exercises?

In the guided exercises, I learned how to test if a system is time-invariant or time-varying. The idea is to shift the input and output and then check if they still match. If they match, the system is time-invariant; if not, it is time-varying. I also learned that systems which depend directly on time usually end up being time-varying. Using MATLAB made it easier to see and understand these results.

From the guided exercises, I observed that some systems react differently when the input is shifted. Time-invariant systems behave consistently, while time-varying ones do not. I also noticed that MATLAB graphs were very helpful in showing these differences, since I could clearly see when the outputs matched or not. This made the concepts easier to understand compared to just reading the theory.

In the problem exercises, I found that some systems were time-varying while others were time-invariant. Problems 1, 2, 3, 5, and 6 were time-varying, while only Problem 4 was time-invariant. This showed me that systems can behave differently, and checking them properly is important. I also saw that time-invariant systems are easier to predict and analyze compared to time-varying ones. Using MATLAB confirmed my answers and made the activity more understandable.